

Learner Demographics and Course Enrollment Behavior Analysis on EduPro

A Descriptive Analytics Research Report

Abstract

This report presents a descriptive analysis of learner demographics and course enrollment behavior on the EduPro online learning platform. Using 3,000 real learner records combined with course and transaction data, the analysis examines age and gender distribution, course category and level popularity, enrollment concentration, and time-based enrollment trends. Key findings show a young, evenly gender-split learner base (ages 15-35, 50.7% female / 49.3% male) with high repeat engagement (69.5% of learners enroll in more than one course) and a moderate concentration effect, where the top 10% of learners account for 28.5% of all enrollments. Marketing and Programming are the leading course categories, and Beginner-level content dominates enrollment volume.

Note: Course and Transaction data used in this analysis are synthetically generated (see Methodology) because the original dataset provided only learner demographic records. Findings involving courses/enrollments should be read as a demonstration of the analytical method, not as verified real-world platform behavior, until real course/transaction data is substituted.

Introduction

Online learning platforms generate large volumes of behavioral data, but that data only becomes useful once it is cleaned, structured, and explored. This project analyzes learner demographics (age, gender) against course enrollment behavior (category, level, timing) to surface patterns that could inform platform strategy — which content to expand, which learner segments are underserved, and how engagement is distributed across the user base.

Problem Statement

EduPro (like many learning platforms) accumulates enrollment data without a clear picture of *who* is enrolling in *what*, or *when*. Without this visibility, content and marketing decisions are made on intuition rather than evidence. This project addresses that gap through structured, descriptive analysis — no predictive modeling is used, keeping the scope focused on understanding the current state of the data.

Objectives

1. Clean and merge the Users, Courses, and Transactions datasets into a single analysis-ready table.
2. Quantify learner demographics (age distribution, gender split, age groups).
3. Quantify enrollment behavior (top/least popular courses, category and level popularity, enrollment concentration).
4. Cross-tabulate demographics against course preferences (age group vs. category, gender vs. level, etc.).
5. Surface time-based enrollment trends (monthly, quarterly, yearly).
6. Translate findings into actionable business and policy recommendations.

Dataset Description

Sheet	Rows	Key Columns	Source
Users	3,000	UserID, UserName, Age, Gender, Email	Real
Courses	40	CourseID, CourseName, CourseCategory, CourseType, CourseLevel	Synthetic
Transactions	9,164	TransactionID, UserID, CourseID, TransactionDate	Synthetic

Synthetic Courses/Transactions were generated with structure rather than pure randomness: course popularity follows a power-law-like distribution (a few courses dominate enrollment, most don't), enrollment counts per learner follow a realistic long-tail (most learners take 1-4 courses, a small "power user" segment takes many more), and category preference is softly correlated with age band, mimicking plausible real-world patterns.

Methodology

Data Cleaning

- Removed duplicate rows and rows with missing primary keys.
- Standardized Gender values (whitespace/casing normalized; invalid values mapped to "Other").
- Removed implausible ages (<5 or >100) and imputed the remainder with the median age.
- Converted `TransactionDate` to proper datetime objects, dropping rows where parsing failed.
- Merged Users → Transactions → Courses on `UserID` and `CourseID` into one master table (9,164 rows, zero missing values after merge).

Feature Engineering

- **Age Groups:** `<18`, `18-25`, `26-35`, `36-45`, `46+`
- **Date parts:** Year, Month, Month Name, Quarter, Day Name — extracted from `TransactionDate` to support all time-trend analysis.

EDA & Visualization

Descriptive statistics and 17 Plotly visualizations (bar, pie/donut, histogram, line, stacked/grouped bar, heatmap, treemap, sunburst, boxplot, scatter) were produced across demographics, enrollment, course, and cross-tabulated demographic-preference views. See the Streamlit dashboard for the interactive versions of every chart referenced below.

Insights

Demographics

- The real learner base spans ages **15-35 only** — no learners aged 36+ appear in the dataset. Age groups: `<18` = 433, `18-25` = 1,121, `26-35` = 1,446, `36-45` = 0, `46+` = 0.
- Gender split is nearly even: **50.7% Female, 49.3% Male**.
- Average learner age: **25.0 years**.

Enrollment Behavior

- Total enrollments: **9,164** across **3,000 learners** (avg. **3.05** courses/learner).
- Top course: *SEO Fundamentals* (968 enrollments), followed by *Adobe Illustrator Essentials* (671) and *Object Oriented Programming* (623).
- Least popular: *Figma Masterclass* (37 enrollments) and *Python for Data Science* (39).
- **69.5%** of learners enrolled in more than one course (strong repeat engagement).
- The top 10% most active learners account for **28.5%** of all enrollments — a moderate concentration effect, not an extreme one.

Course Preferences

- **Marketing** is the leading category (2,567 enrollments), followed by Programming (1,273) and Personal Development (1,257). Cloud Computing is least popular (599).
- **Beginner**-level courses dominate (5,244 enrollments, 57% of total), followed by Intermediate (3,282) and Advanced (638).
- **Paid** courses outnumber Free ones roughly 2.4:1 (6,440 vs. 2,724).

Time Trend

- Enrollment grew from **3,492** (2024) to **5,672** (2025) — a ~62% year-over-year increase, consistent with the mild upward trend built into the synthetic transaction dates.

Business Recommendations

- Prioritize content investment in **Marketing and Programming**, the two highest-enrollment categories.
- Since Beginner-level content drives the majority of volume, consider building clearer **Beginner → Intermediate → Advanced** learning paths to convert existing engagement into higher-value course completions.
- Target the **repeat-learner segment (69.5%)** with bundle or subscription offers, since they are already demonstrating multi-course intent.
- Investigate low-performing courses (*Figma Masterclass*, *Python for Data Science*) for pricing, marketing, or content-quality issues before assuming low intrinsic demand.

Government / Policy Recommendations

(Speculative — contingent on this pattern reflecting a real regional population once real data is used.)

- The complete absence of learners aged 36+ may indicate a **digital upskilling gap** for older workers; targeted digital-literacy or reskilling outreach could help close it.
- Public subsidy or grant programs could focus on **increasing Advanced- level course access**, since Advanced content is currently the smallest segment (638 enrollments, ~7% of total) despite likely economic value.

Conclusion

This analysis demonstrates a complete descriptive-analytics pipeline — cleaning, merging, feature engineering, EDA, and visualization — applied to learner demographic and enrollment data. Even

with synthetic course/transaction data, the pipeline surfaces coherent, actionable patterns: a young and gender-balanced learner base, strong repeat engagement, a moderate concentration of activity among top learners, and clear category/level preferences. The modular codebase (`loader.py` , `preprocess.py` , `metrics.py` , `charts.py`) is structured so that swapping in real Courses/Transactions data requires no changes to the analysis logic itself.

Future Scope

- Replace synthetic Courses/Transactions data with real platform data.
- Add cohort and retention analysis (do learners return month-over-month?).
- Incorporate course completion rates and learner ratings, not just enrollment counts, to distinguish engagement from genuine value delivery.
- Extend the age range check once real data with a broader age spread is available, to confirm whether the 36+ gap is a data artifact or a real platform pattern.